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Roll no.

Degree: B.Tech. Course work
END-SEMESTER EXAMINATION (SUMMER), July 2026

Course Title: Basics of Analog and Digital Electronics
Course Code: FCEC0116

Duration: 3:00 Hours

Max. Marks: 50

Note: - Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
1	Attempt any 2 parts of the following		
1a	A half-wave rectifier is supplied with an input voltage of 24 V (rms). Calculate the peak output voltage, average DC output voltage, ripple factor, and rectification efficiency.	5	1
1b	Differentiate between a clipper and a clamper circuit. Explain their operating principles and practical applications with suitable diagrams.	5	1
1c	Differentiate between a CE amplifier and a CS amplifier with respect to gain, input impedance, output impedance, phase shift, and applications.	5	1
2	Attempt any 2 parts of the following		
2a	Compare an integrator and a differentiator using operational amplifiers with respect to circuit configuration, output waveform, frequency response, and applications.	5	2
2b	An op-amp has a slew rate of 0.5 V/ μ s and a gain-bandwidth product of 1 MHz. Can it amplify a 100 kHz sine wave with a peak output of 10 V without distortion? Justify your answer.	5	2
2c	An inverting amplifier is designed using an operational amplifier with an input resistor of 5 k Ω and a feedback resistor of 100 k Ω . If the input voltage is 0.8 V and the op-amp is powered by ± 15 V, calculate the closed-loop voltage gain, the output voltage, and whether the amplifier operates in the linear region or enters saturation.	5	2
3	Attempt any 2 parts of the following		
3a	Explain the working of the IC 555 timer as a monostable multivibrator with a neat circuit diagram. Derive the expression for the pulse width.	5	3
3b	State and explain the Barkhausen criterion for oscillation. Describe the working of an RC phase-shift oscillator using an operational amplifier.	5	3
3c	Compare the RC phase-shift oscillator and Wien bridge oscillator with respect to circuit configuration, working principle, frequency range, and applications.	5	3
4	Attempt any 2 parts of the following		
4a	Explain the concept of universal gates and implement the NOT, OR, and AND gates using only NOR gates.	5	4

4b	Explain the working of a 4-bit Parallel-In Serial-Out (PISO) shift register with a suitable diagram. List the applications of shift registers. Calculate the number of clock pulses required for parallel loading and serial output of 4-bit data.	5	4
4c	Explain the working of a MOD-10 ripple counter with a suitable diagram. Discuss the decoding errors in ripple counters and suggest methods to eliminate them.	5	4
5	Attempt any 2 parts of the following		
5a	Draw the block diagram of the 8085 microprocessor and explain the functions of its major functional blocks. Also list any four applications of the 8085 microprocessor.	5	5
5b	Write an 8085 assembly language program to add two 8-bit numbers and store the result in memory. Differentiate between a programmable processor and a microcontroller by giving any four points.	5	5
5c	Write 8085 assembly language programs to perform the logical AND, OR, and XOR operations on two 8-bit numbers stored in memory locations. Store each result in a separate memory location.	5	5